Block Cross-Column Krylov Gram Certificates

Preserving Packet Phases in Weighted Stein Tails

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Abstract

RH-65 showed that a global Lyapunov metric cannot replace peripheral-mode removal. The remaining route is to project first and weight only the final localized residual. Existing RH-62–RH-64 certificates were vector-valued, however, while the phase-aware Schur construction of RH-58–RH-60 is a cross-column Gram problem. Applying a positive upper to each packet column separately destroys the cancellation that made the finite-horizon route effective.

This paper gives a block remedy. For an isometry V, block source $Z = VB+E$, projected matrix $H = V^*AV$, and Galerkin residual $R = AV - VH$, we prove the exact identity

$$A^L Z = VH^L B + \sum_{j=0}^{L-1} A^{L-1-j} RH^j B + A^L E.$$

In a positive metric with contraction $q < 1$, the identity yields (i) a coefficient-aware center-radius certificate retaining all packet phases and (ii) a positive-semidefinite Gram envelope valid simultaneously for every coefficient vector. Both use the cross-column matrices $(H^j B)^* R^* MR(H^j B)$ before positive majorization.

On a two-column slow-mode cancellation witness at horizon 32, independent column bounds lose a factor 2.129×10^{18}, whereas the block directional certificate has gain 1.0000014. A 256-bit Arb audit certifies that the independent-column loss exceeds 10^{18}. On a nonnormal four-step chain, one block level reduces the gain from 11.37 to 3.90 and rank-four closure is exact. A six-mode complex-phase model improves from 1.713 to 1.288 before exact rank-six closure. The uniform PSD envelope still has gain 410 in the specially cancelling direction, exposing the next gate: coefficient-adapted positive residual geometry. No production block depth theorem, Stage A1 closure, arithmetic trace formula, or Hilbert–Polya conclusion is claimed.

Keywords: block Krylov method; cross Gramian; phase cancellation; Lyapunov metric; Stein tail; packet fusion.

MSC 2020: 47A10; 47B65; 65F35; 93B07; 93D05.

1 Introduction

The directional Hardy route is intrinsically multi-source. Schur packets produce columns $z_1, \dots, z_r$ and a coefficient vector a; the physical quantity is the norm of the coherent sum Za, not the sum of the packet norms. RH-58 kept this information in a time-ordered cross Gram, while RH-60 retained it over a finite horizon. The terminal estimates of RH-62–RH-64 then returned to one source vector [3, 4].

RH-65 determines the proper order: peripheral or reached slow directions must be removed before a terminal metric is applied [2]. To transfer that order to packets, the projection must be block-valued. Block Krylov spaces are standard for multiple right-hand sides [1]; the point here is a specific positive Gram certificate that does not scalarize the columns prematurely.

Contributions and boundary

1. We prove a block Galerkin power identity with an explicit source reconstruction residual.
2. We derive a directional center-radius certificate and a uniform PSD Gram envelope in a Lyapunov metric.
3. We identify exact cancellation and invariant-block closure criteria.
4. We audit three finite models and isolate the gap between a selected physical direction and one envelope valid for all directions.

All matrix theorems below are finite-dimensional. The model calculations are deterministic binary64 evidence; only the displayed cancellation witness has a 256-bit interval audit.

2 Block power identity

Let $A \in \mathbb{C}^{n\times n}$, $Z \in \mathbb{C}^{n\times r}$, and let $V \in \mathbb{C}^{n\times k}$ satisfy $V^*V = I$. Define

$$B = V^*Z, \qquad E = Z - VB, \qquad H = V^*AV, \qquad R = AV - VH. \tag{1}$$

For a block Krylov basis, V spans a numerical approximation to $\operatorname{span}\{Z, AZ, \ldots, A^{d-1}Z\}$, but the next identity does not require that construction.

Proposition 2.1 (Block Galerkin power identity)**.** *For every integer $L \geq 0$,*

$$A^L Z = Y_L + \mathcal{R}_L, \qquad Y_L = VH^L B, \tag{2}$$

where

$$\mathcal{R}_L = \sum_{j=0}^{L-1} A^{L-1-j} R H^j B + A^L E. \tag{3}$$

Proof. The relation $AV = VH + R$ implies by induction

$$A^L VB = VH^L B + \sum_{j=0}^{L-1} A^{L-1-j} R H^j B.$$

Add $A^L E$ and use $Z = VB + E$. ☐

The identity keeps every source column inside B and $H^j B$. No triangle inequality has yet been applied across packets.

3 Cross-column weighted certificates

Let $M \succ 0$ and suppose

$$q = \left\| M^{1/2} A M^{-1/2} \right\|_2 < 1. \tag{4}$$

The canonical equation $M - A^*MA = I$ is one sufficient construction [5]. Put

$$Q = R^*MR, qquad Q_E = E^*ME, qquad C_j = H^j B. \tag{5}$$

Theorem 3.1 (Directional block center-radius certificate)**.** *For every $a \in \mathbb{C}^r$,*

$$\left\|A^L Z a\right\|_M \le \|Y_L a\|_M + \rho_L(a), \tag{6}$$

where

$$\rho_L(a) = \sum_{j=0}^{L-1} q^{L-1-j}\sqrt{a^* C_j^* Q C_j a} + q^L\sqrt{a^* Q_E a}. \tag{7}$$

Consequently

$$a^*(A^L Z)^* M (A^L Z) a \le \left(\sqrt{a^* Y_L^* M Y_L a} + \rho_L(a)\right)^2. \tag{8}$$

Proof. Apply (4) to each propagated term in (3). Since $\|RC_j a\|_M^2 = a^* C_j^* Q C_j a$, the triangle inequality gives (6); squaring gives (8). □

For a certificate valid for every coefficient simultaneously, choose positive weights satisfying

$$\theta_0 + \cdots + \theta_{L-1} + \theta_E = 1,$$

and define

$$\mathcal{C}_L(\theta) = \sum_{j=0}^{L-1} \frac{q^{2(L-1-j)}}{\theta_j} C_j^* Q C_j + \frac{q^{2L}}{\theta_E} Q_E. \tag{9}$$

Theorem 3.2 (Uniform PSD Gram envelope)**.** *The residual and full block Grams obey*

$$\mathcal{R}_L^* M \mathcal{R}_L \preceq \mathcal{C}_L(\theta), \tag{10}$$

$$(A^L Z)^* M (A^L Z) \preceq (1+\eta) Y_L^* M Y_L + (1+\eta^{-1})\mathcal{C}_L(\theta) \tag{11}$$

for every $\eta > 0$.

Proof. For a fixed a, apply the weighted Cauchy inequality $(\sum_i x_i)^2 \le \sum_i x_i^2/\theta_i$ to the summands in (7). This proves $\|\mathcal{R}_L a\|_M^2 \le a^* \mathcal{C}_L(\theta) a$ for every a, which is (10). The operator Young inequality

$$(X+W)^*(X+W) \preceq (1+\eta) X^* X + (1+\eta^{-1}) W^* W$$

with $X = M^{1/2} Y_L$ and $W = M^{1/2}\mathcal{R}_L$ then gives (11). □

The implementation chooses trace-optimal scalar weights:

$$\theta_j \propto q^{L-1-j}\sqrt{\operatorname{tr}(C_j^* Q C_j)}, \qquad \theta_E \propto q^L \sqrt{\operatorname{tr} Q_E}, \tag{12}$$

and then $\eta = \sqrt{\operatorname{tr}\mathcal{C}_L / \operatorname{tr}(Y_L^* M Y_L)}$. These choices minimize the traces of the two successive positive envelopes. They need not be optimal for a selected coefficient direction.

Corollary 3.3 (Exact fused cancellation)**.** *If a coefficient vector a satisfies*

$$Ea = 0, qquad RH^j Ba = 0 \quad (0 \le j < L), \tag{13}$$

then $\rho_L(a) = 0$ and the directional certificate is exact. This may hold even when none of the individual source columns has zero residual.

Proof. Every quadratic term in (7) vanishes. □

If $E = R = 0$, the block subspace is invariant and both certificates are exact for all coefficient vectors.

4 Finite-model audit

The code builds V by an SVD of the block Krylov matrix and uses the canonical Lyapunov metric. Four uppers are compared: the directional block certificate, the uniform PSD Gram evaluated in the physical direction, independent scalar-column certificates followed by Minkowski, and a rank-matched Krylov certificate built only after fusing the selected source.

Table 1: One-level directional energy gains.

model	rank	block dir.	PSD Gram	independent	fused
cancelling slow pair	2	1.000001	410.159	2.129×10^{18}	1.000000
nonnormal chain	3	3.899145	3.950049	11.3650	5.007690
complex phase packets	3	1.288471	1.295052	1.71263	1.129739

The cancellation witness has

$$A = \mathrm{diag}(0.995, 0.55, 0.2), \qquad Z = \begin{pmatrix} 1 & -1 \\ 1 & 1 \\ 0.2 & -0.2 \end{pmatrix}, \qquad a = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Thus $Za = 2e_2$: the slow and fast components cancel before propagation. The block space contains the invariant fused direction and satisfies Corollary 3.3; floating-point orthogonalization leaves only a 1.4×10^{-6} relative gain. Separate columns must each carry the 0.995 mode and therefore lose more than eighteen orders of magnitude. A 256-bit Arb calculation certifies the cancellation, positivity of the exact fused energy, residual annihilation on the fused axis, and an independent- column lower loss exceeding 10^{18}.

The same example reveals a distinction. The trace-global PSD envelope must also cover coefficient vectors that do not cancel the slow mode, so its evaluation at $a = (1, 1)$ still has gain 410. The theorem is valid; the choice of one global positive envelope is geometrically mismatched to a special physical direction.

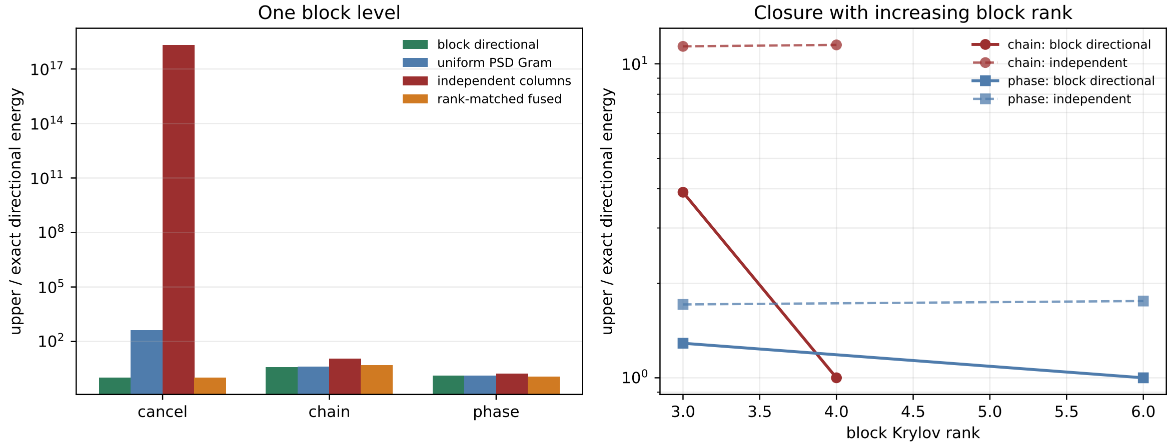

Figure 1: Left: one-level gains, including the catastrophic independent-column loss in the cancellation witness. Right: increasing block rank closes the nonnormal and complex-phase models, while independent scalarization retains a packet-fusion penalty.

For the nonnormal chain, rank three improves the directional gain from the independent value 11.37 to 3.90; rank four is exact. For the six-mode phase model, rank three improves 1.713 to 1.288 and rank six is exact. The minimum eigenvalues of the numerical PSD slack are nonnegative up to 4.4×10^{-18} rounding.

5 Route consequence

RH-66 repairs the cross-column scalarization loss. The viable terminal architecture is now

$$\text{block Krylov center} \longrightarrow \text{coherent physical coefficient} \longrightarrow \text{localized weighted residual.} \tag{14}$$

For a fixed physical coefficient, Theorem 3.1 can be nearly exact even when every columnwise estimate fails.

The unresolved question is stronger: Stage A1 ultimately needs a positive object that can be transferred through packet and continuum limits. The trace-global Young envelope in Theorem 3.2 can overpay for coefficient directions irrelevant to the physical fusion. The next gate is therefore a weighted block residual stress test comparing directional, low-rank cone, and full PSD formulations. It must determine how much positivity is truly needed by the downstream determinant argument.

6 No arithmetic or Hilbert–Polya conclusion

This paper constructs no self-adjoint operator, no $T \log T$ counting law, no prime-power trace formula, and no completed-zeta identity. It makes no Hilbert–Polya or Riemann-hypothesis claim. Stage A1 and unconditional Stage A4 remain open.

7 Reproducibility

The directory contains block algebra, tests, the model pilot, Arb audit, figures, hashes, and publication artifacts. The principal commands are:

```
pytest -q -p no:cacheprovider
python experiments/run_block_gram_pilot.py
python experiments/run_arb_cancellation_audit.py
MPLBACKEND=Agg python experiments/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The block identities and positive envelopes are analytic. The model gains are binary64 evidence. Uniform physical-family block depth, interval- certified production packet transfer, Stage A1, unconditional Stage A4, a self-adjoint Hilbert–Polya operator, an arithmetic trace formula, and a zeta-zero identity remain open.

References

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